Learning Fine Motor Skills in Virtual Reality: A Comparative Study of Muscle Memory Acquisition

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ACM Reference Format:
Riley Nelson. 2026. Learning Fine Motor Skills in Virtual Reality: A Comparative Study of Muscle Memory Acquisition. In . ACM, New York, NY, USA, **??** pages. magentahttps://doi.org/10.1145/nnnnnnn.nnnnnnn

Abstract

Muscle memory underpins a wide range of human activities, from daily routines to high-stakes professional skills. Virtual Reality (VR) offers a promising medium for motor skill training by providing immersive, low-risk practice environments. This paper describes an in-progress study that compares fine motor muscle memory acquisition in a VR setting versus equivalent real-world practice. Participants learn a pattern-drawing task—inspired by smartphone pattern-lock interaction—in either a VR classroom environment or a physical classroom, then are assessed on accuracy and completion time in a final real-world test. The work contributes to the field of human-computer interaction by examining how interface design, controller input, and immersive visual feedback influence motor skill learning.

CCS Concepts

• **Human-centered computing → Human computer interaction (HCI)**; **Virtual reality**.

Despite growing research interest in VR-based motor learning, relatively few studies directly compare skill acquisition in immersive VR with equivalent real-world practice on identical fine motor tasks. This gap motivates the present study, which asks: *Can participants learn a fine motor pattern-drawing task as quickly and as*

accurately in Virtual Reality as they can in a real-world setting, and does that learning transfer to real-world performance?

To answer this question, we designed a controlled experiment in which participants in two groups—one practicing in VR, one practicing in-person—learn to replicate a dot-connection pattern analogous to a smartphone pattern-lock gesture. Both groups watch the same instructional video, practice the pattern to criterion, and then complete a timed, real-world transfer test. The dependent measures are accuracy and completion time on the final test.

This work sits at the intersection of VR interface design and motor learning research. By choosing a task that maps naturally from physical tools (whiteboard marker) to VR controllers, and by embedding the task in an immersive classroom environment, we aim to isolate the effect of the learning medium while minimizing confounds introduced by unfamiliar input devices or disorienting virtual environments.

2 Related Work

2.1 Virtual Reality in Motor Skill Training

The application of VR to motor skill training is a well-established research area despite the relative youth of the technology. Kampiotis and Theodorakou **?**,]]kampiotis2003 conducted a wide-ranging literature review documenting VR training applications spanning military physicians, naval operators, and automobile drivers, noting that early VR training systems appeared as early as the 1990s in contexts such as pilot simulation. Their review concluded that VR's interactive and immersive properties make it particularly well-suited to the acquisition of motor skills, where active engagement with the task is essential.

More recent empirical work reinforces this view. Pastel et al. **?**,]]pastel2022 demonstrated that training in VR enables the acquisition of a complex sports movement, showing performance gains comparable to those seen in real-world practice conditions. This finding is significant because it suggests that the perceptual richness of VR—including stereoscopic depth, spatial audio, and real-time haptic feedback from controllers—can provide the sensory information needed to drive motor program formation.

2.2 Input Devices and Sensory Fidelity

A recurring theme in VR training research is the importance of input device selection. Hardware such as head-mounted displays (HMDs), body-worn motion capture suits, and data gloves can dramatically enrich the sensory fidelity of a VR experience. Data gloves in particular are noted as a promising extension for fine motor tasks, because they capture individual finger movements and can provide force feedback that approximates the resistance felt when grasping a physical tool **?**,]]kampiotis2003. However, access to such specialized peripherals is often limited in educational or resource-constrained research settings. The present study uses standard hand-held controllers, which, while less expressive than data gloves, nonetheless approximate the gross motor demands of holding and guiding a whiteboard marker and are widely available.

2.3 Mixed Reality Displays and Observational Learning

Watson and Livingston **?**,]]watson2018 examined how different mixed reality display types—including VR—affect observational learning of motor skills. Their design-research study highlighted several critical factors for effective VR motor training: (1) the importance of ensuring that technological interfaces do not interrupt or distract from the learning experience itself; (2) the need to present movement demonstrations in ways that minimize breaks in immersion; and (3) the value of embedding modeling behavior within the virtual environment rather than relying on external instruction. These design principles directly informed the present study. In particular, the decision to project the demonstration video within the VR classroom—rather than pausing the VR session to show an external screen—reflects Watson and Livingston's guidance on maintaining immersive continuity.

2.4 VR in Neurorehabilitation and Skill Transfer

Beyond healthy population training, VR has been applied in clinical contexts to support motor recovery. McDermott et al. **?**,]]mcdermott2021 explored the use of EEG-informed VR pipelines for neurorehabilitation, illustrating that VR environments can be tightly coupled with physiological monitoring to detect motor-relevant brain states. While this line of work operates at a different scale of complexity than the present study, it underscores the broad applicability of VR-based motor interventions and raises the possibility that VR training environments may engage neural motor circuits in ways that support durable learning.

2.5 Research Gap

Taken together, the literature demonstrates that VR is an effective platform for motor skill training across a range of domains and populations. However, fewer studies directly compare *equivalent* fine motor tasks performed in VR versus the real world, controlling for the same stimulus material, the same task structure, and the same real-world transfer test. The present study addresses this gap by using a task—dot-pattern drawing—that is structurally identical across both conditions, differing only in the medium through which it is practiced.

3 Methodology

3.1 Study Design

This study employs a between-subjects design with two conditions: a *VR training condition* and a *real-world training condition*. Participants are randomly assigned to one condition at the outset of the experiment. The independent variable is the training environment (VR vs. real-world). The primary dependent variables are (1) accuracy of pattern replication in the final transfer test, operationalized as a binary correct/incorrect judgment, and (2) time to complete the pattern in the final transfer test.

3.2 Task

The motor task is a pattern-drawing exercise modeled on the pattern-lock interaction found on smartphones. A grid of dots is presented on a vertical surface (a whiteboard in both conditions), and participants must connect the dots in a specific sequence using straight lines. The pattern is designed to be moderately complex—requiring several non-obvious direction changes—so that it cannot be memorized trivially after a single viewing, but is learnable within a short practice session. The use of a vertical whiteboard surface was chosen for two reasons: it aligns naturally with VR interaction, where drawing downward onto a horizontal surface is awkward with current controller setups, and it can be replicated faithfully in the real-world condition using a physical whiteboard.

3.3 Virtual Reality Environment

The VR environment is implemented in Unity and deployed on a Meta Quest 3 head-mounted display. The virtual scene represents a classroom: the participant stands (or sits) facing a projection screen on which the demonstration video plays, and a whiteboard to one side displays the dot grid. Controller input is used to draw lines connecting the dots, with a line-snapping mechanism implemented to make the interaction feel fluid and to reduce the motor load attributable solely to controller manipulation. Snap-to-dot assistance ensures that precision demands are comparable across the VR and real-world conditions. A future enhancement under consideration is automated pattern recognition within the VR environment, which would provide the participant with immediate correctness feedback without requiring experimenter intervention, thereby enabling a fully self-guided VR experience.

3.4 Real-World Condition

Participants in the real-world condition view the same demonstration video on a physical screen. The dot grid is pre-printed on a physical whiteboard using permanent marker, and participants draw the pattern using a standard whiteboard marker. The experimenter manages session pacing—signaling trial start and stop—using the same timing structure as the VR condition, ensuring procedural equivalence between groups.

3.5 Procedure

Both conditions follow an identical procedural sequence:

(1) **Observation.** The participant watches the demonstration video showing the target pattern being drawn.
(2) **Practice trials.** The participant attempts to replicate the pattern. After each failed attempt, the participant re-watches the video and verbally reports what elements of the pattern they missed. Practice continues until the participant successfully reproduces the pattern.
(3) **Timed repetitions.** Once criterion is reached, the participant completes three timed trials to allow within-session consolidation and to expose any speed-accuracy tradeoffs.
(4) **Transfer test.** Both groups complete a final timed trial on paper and pencil in the real world (i.e., the VR group switches to paper for this test). This transfer test is the primary data collection point.

The demonstration video is held constant across conditions, ensuring that any differences in learning outcomes can be attributed to the practice medium rather than to differences in instructional content.

3.6 Measures and Analysis

Transfer test performance is assessed along two dimensions:

- **Accuracy:** A binary score indicating whether the participant reproduced the pattern correctly or incorrectly.
- **Completion time:** The elapsed time from trial start signal to final pen lift, recorded in seconds.

Group differences in accuracy will be analyzed using a chi-square test of independence. Group differences in completion time (for correct trials) will be analyzed using an independent-samples t-test or a Mann-Whitney U test, depending on the distributional properties of the data. Results will be compared against findings from the broader VR motor learning literature to situate any observed effects in the context of prior work.

3.7 Participant Assignment and Controls

Because pattern complexity is a potential confound—some patterns may be intrinsically harder to memorize than others—both groups learn the same pattern, eliminating pattern-difficulty as a between-group variable. Participants are assigned to groups with the goal of achieving balanced group sizes. Potential confounds to be documented include prior VR experience (which may ease or hinder learning in the VR condition) and prior pattern-lock usage on mobile devices (which may give some participants a head start on the task structure). Participants will be surveyed on both dimensions at study intake.